

Ali feels that battery and power management weren't a concern for embedded engineers when it came to applications based on primary (non-rechargeable) electrical storage. But in devices using secondary (rechargeable) electrical storage, "Multiple user-interface and transactional requirements of the various components and modules, and the overall performance and operational expectations of play-time from a user's perspective have made it compelling to understand the battery," he says.

Based on their construction, Li-ion batteries can be classified as cylindrical and prismatic in shape. Further differentiation is in the inner anode and cathode material which could be layered in flat stacks or a long coiled layout. These formats reflect the selection of the cell type based on the application. The classification is also done based on the type of electrolyte used (lithium-iron phosphate, lithium polymer) and the combination of rare elements (magnesium, nickel, cobalt,

### Some Li-ion battery vendors

- BAK
- BYD
- Henan Huanyu
- ENIX
- Maxell
- Panasonic
- Renata
- Sony
- Varta
- Valence

etc), which can vary in performance and price.

Ali says, "The refined nature of the chemicals used plays an important role."

The major disadvantages of Li-ion batteries are combustion due to overheating and leakage. Also, very deep discharge may result in short-circuit. To avoid such a situation, the Li-ion battery uses a fail-safe circuitry which shuts down the cell outside the safe range of 3-4.2V.

### Let your battery last longer

The key reason for the fading performance of a battery is the chemistry of

the cell. Degradation is bound to happen in any chemical composition but it can certainly be controlled by improving the way the battery is used.

Michael Lin, material scientist, Quallion, shares, "The cell ages faster if you push it to more extreme states. If there is a higher state of charge and higher depth of discharge, which means you are operating your battery over a large voltage range, then the ageing is faster. In the case of high-power input/output, there is larger polarisation and the voltage becomes too high/too low while charging or discharging again, leading to faster ageing. Also, very high temperature leads to faster self-discharge, while very low temperature leads to large polarisation, which again degrades the performance of the cell."

But ultimately, it depends on the cell chemistry that you use in your battery.

The best way to monitor ageing is to look at the growth of internal resistance. Neal Golovin, battery applications



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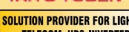












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